

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_no_replay.
- Best TSPLIB holdout gap: phase6_no_replay.
- Best synthetic holdout gap: phase6_stratified_random_replay.

Run Metadata

- run_name: run_20260518_235218_e
- started_at_local: 2026-05-18 23:52:18
- finished_at_local: 2026-05-19 00:11:46
- duration_hhmm: 00:19
- duration_seconds: 1168.327
- seed_offset: 4000
- replicate_label: e
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.081748	0.002656	0.052987	0.236772	0.160079	0.0	0.788654	0.76	0.046072
phase6_random_replay	random	score_only	False	0.135463	0.001628	0.086796	0.236772	0.168557	0.19457	0.706409	0.76	0.062175
phase6_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.171206	0.348017	0.261637	0.880734	0.56	-0.028278

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.213387	0.064916	0.159398	0.236772	0.214847	0.19457	0.0	0.56	0.0
phase6_stratified_random_replay	stratified_random	score_only	False	0.091054	0.0	0.057943	0.236772	0.15932	0.221162	0.625641	0.76	0.053
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.151349	0.001628	0.096905	0.236772	0.165767	0.130039	0.604411	0.76	0.052267
phase6_residual_failure_replay	residual_failure	score_only	False	0.213387	0.064916	0.159398	0.149274	0.338865	0.291666	0.901432	0.56	-0.027628
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.127487	0.0	0.081128	0.171206	0.265853	0.171783	0.836498	0.76	0.019349
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.20855	0.0	0.132714	0.171206	0.284075	0.173315	0.0	0.76	0.0

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.081748, synthetic 0.002656, combined 0.052987.
- Accepted-epoch count 3, mean accepted code novelty 0.788654, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.160079, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency 0.046072 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.081748 across 7 instances; family means: ch=0.057102, kroD=0.074199, pcb=0.210741, pr=0.055816, rd=0.041719, st=0.075556.
- Panel synthetic_holdout mean gap 0.002656 across 4 instances; family means: clustered_gaussian=0.010624, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3364, "expected_gap": 0.404343, "family": "a", "name": "a280", "optimality_gap": 0.304382, "residual_gap": -0.099961}, {"best_known_cost": 14379, "cost": 17686, "expected_gap": 0.326838, "family": "lin", "name": "lin105", "optimality_gap": 0.229988, "residual_gap": -0.09685}],

{"best_known_cost": 629, "cost": 715, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.136725, "residual_gap": -0.120827}}

phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.135463, synthetic 0.001628, combined 0.086796.
- Accepted-epoch count 2, mean accepted code novelty 0.706409, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.168557, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133995, mean selected residual gap 0.007389.
- Adaptation efficiency 0.062175 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.135463 across 7 instances; family means: ch=0.106939, kroD=0.071757, pcb=0.2365, pr=0.177683, rd=0.143236, st=0.105185.
- Panel synthetic_holdout mean gap 0.001628 across 4 instances; family means: clustered_gaussian=0.006511, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 7542, "cost": 9897, "expected_gap": 0.245028, "family": "berlin", "name": "berlin52", "optimality_gap": 0.312251, "residual_gap": 0.067223}, {"best_known_cost": 14379, "cost": 18034, "expected_gap": 0.326977, "family": "lin", "name": "lin105", "optimality_gap": 0.25419, "residual_gap": -0.072787}, {"best_known_cost": 2579, "cost": 3157, "expected_gap": 0.40442, "family": "a", "name": "a280", "optimality_gap": 0.224118, "residual_gap": -0.180302}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.880734, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.261637.
- Replay selection diversity 0.177669, mean selected expected gap 0.309513, mean selected residual gap 0.066244.
- Adaptation efficiency -0.028278 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.406824, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.005506}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.25469, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.136407}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.32496, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.003853}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132191, mean selected residual gap 0.039175.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.404498, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.00318}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.253418, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.137679}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.321761, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.007052}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.091054, synthetic 0.0, combined 0.057943.
- Accepted-epoch count 3, mean accepted code novelty 0.625641, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.15932, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.236082, mean selected residual gap -0.013045.
- Adaptation efficiency 0.053 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.091054 across 7 instances; family means: ch=0.055346, kroD=0.069926, pcb=0.185671, pr=0.053782, rd=0.143236, st=0.074074.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3572, "expected_gap": 0.405661, "family": "a", "name": "a280", "optimality_gap": 0.385033, "residual_gap": -0.020628}, {"best_known_cost": 629, "cost": 753, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.197138, "residual_gap": -0.059142}, {"best_known_cost": 14379, "cost": 16519, "expected_gap": 0.321274, "family": "lin", "name": "lin105", "optimality_gap": 0.148828, "residual_gap": -0.172446}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.151349, synthetic 0.001628, combined 0.096905.
- Accepted-epoch count 2, mean accepted code novelty 0.604411, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.165767, size bias 8.3e-05, and failure concentration 0.130039.
- Replay selection diversity 0.263853, mean selected expected gap 0.167628, mean selected residual gap -0.017093.

- Adaptation efficiency 0.052267 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.151349 across 7 instances; family means: ch=0.135874, kroD=0.185029, pcb=0.226771, pr=0.14021, rd=0.142351, st=0.093333.
- Panel synthetic_holdout mean gap 0.001628 across 4 instances; family means: clustered_gaussian=0.006511, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 19809, "expected_gap": 0.323194, "family": "lin", "name": "lin105", "optimality_gap": 0.377634, "residual_gap": 0.05444}, {"best_known_cost": 2579, "cost": 3336, "expected_gap": 0.404188, "family": "a", "name": "a280", "optimality_gap": 0.293525, "residual_gap": -0.110663}, {"best_known_cost": 629, "cost": 717, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.139905, "residual_gap": -0.114149}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.901432, and final complexity 0.56.
- Active replay archive residual_archive with mean diversity 0.149274, mean hardness 0.338865, size bias 0.093558, and failure concentration 0.291666.
- Replay selection diversity 0.177489, mean selected expected gap 0.264849, mean selected residual gap 0.087544.
- Adaptation efficiency -0.027628 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.405196, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.003878}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.129094}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.322373, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.00644}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.127487, synthetic 0.0, combined 0.081128.
- Accepted-epoch count 3, mean accepted code novelty 0.836498, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.265853, size bias 0.120322, and failure concentration 0.171783.
- Replay selection diversity 0.198691, mean selected expected gap 0.299333, mean selected residual gap -0.032853.
- Adaptation efficiency 0.019349 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.127487 across 7 instances; family means: ch=0.120487, kroD=0.142012, pcb=0.220568, pr=0.112048, rd=0.142731, st=0.034074.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3117, "expected_gap": 0.403955, "family": "a", "name": "a280", "optimality_gap": 0.208608, "residual_gap": -0.195347}, {"best_known_cost": 629, "cost": 712, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.131955, "residual_gap": -0.135136}, {"best_known_cost": 14379, "cost": 16239, "expected_gap": 0.321803, "family": "lin", "name": "lin105", "optimality_gap": 0.129355, "residual_gap": -0.192448}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.20855, synthetic 0.0, combined 0.132714.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.284075, size bias 0.120322, and failure concentration 0.173315.
- Replay selection diversity 0.196315, mean selected expected gap 0.309713, mean selected residual gap -0.023486.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.20855 across 7 instances; family means: ch=0.174542, kroD=0.343853, pcb=0.231006, pr=0.319779, rd=0.161315, st=0.054815.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3539, "expected_gap": 0.407212, "family": "a", "name": "a280", "optimality_gap": 0.372237, "residual_gap": -0.034975}, {"best_known_cost": 7542, "cost": 10190, "expected_gap": 0.235826, "family": "berlin", "name": "berlin52", "optimality_gap": 0.351101, "residual_gap": 0.115275}, {"best_known_cost": 14379, "cost": 18028, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.253773, "residual_gap": -0.067738}]

Judge Appendix

Analysis of Phase 6 TSP Benchmark Suite

Overview

- 9 conditions tested with various replay mechanisms.
- Main metrics: held-out TSPLIB gap and synthetic holdout gap (transfer evidence).
- Key replay mechanisms: no replay, random replay, failure replay, random replay with compression, stratified random replay, diversity-weighted replay, residual failure replay, diversity failure replay, diversity failure replay with compression.
- Focus on optimality gaps and transfer metrics.

Summary Table of Key Metrics

Condition	Replay Mode	Selection Mode	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Replay Failure Concentration	Code Novelty (mean)
phase6_no_replay	none	score_only	0.0817	0.0027	0.0530	0.2368	0.1601	0.0	0.789
phase6_random_replay	random	score_only	0.1355	0.0016	0.0868	0.2368	0.1686	0.1946	0.706
phase6_failure_replay	failure	score_only	0.2134	0.0649	0.1594	0.1712	0.3480	0.262	0.881
phase6_random_replay_compression	random	novelty_gate	0.2134	0.0649	0.1594	0.2368	0.2148	0.195	0.000
phase6_stratified_random_replay	stratified_random	score_only	0.0911	0.0000	0.0579	0.2368	0.1593	0.221	0.626
phase6_diversity_weighted_replay	diversity_weighted	score_only	0.1513	0.0016	0.0969	0.2368	0.1658	0.130	0.604
phase6_residual_failure_replay	residual_failure	score_only	0.2134	0.0649	0.1594	0.1493	0.3389	0.292	0.901
phase6_diversity_failure_replay	diversity_failure	score_only	0.1275	0.0000	0.0811	0.1712	0.2659	0.172	0.836
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	0.2086	0.0000	0.1327	0.1712	0.2841	0.173	0.000

Optimality and Transfer Performance

- **Best held-out TSPLIB gap and transfer gap:**
- **phase6_no_replay (no replay):** TSPLIB gap 8.17%, transfer gap 5.30%
- **phase6_stratified_random_replay:** synthetic gap is extremely low (0.0), TSPLIB gap improved (9.11%), transfer gap 5.79%
- **Worst transfer and TSPLIB gaps observed for:**
- **phase6_failure_replay** and **phase6_random_replay_compression**: TSPLIB gap ~21%, transfer gap ~16%
- Failure/concentration replay modes have higher gaps.
- **Random replay with score_only selection (phase6_random_replay) shows higher gaps than no replay.**
- **Stratified random replay yields a strong synthetic holdout gap (0.0), near-best TSPLIB and transfer gaps, consistent transfer improvement over baseline.**
- Code novelty is **zero** in compression-novelty-gated replay conditions but high in others.

Replay Mechanism Insights

1. Replay Archive Diversity

- Diversity remains high (~0.23-0.24) for non-failure replay modes.
- Failure replay modes (failure, residual_failure, diversity_failure) show reduced final archive diversity (~0.15-0.17).
- Compression pressure reduces code novelty drastically but maintains archive descriptor diversity in some cases (~0.17-0.23).

2. Hardness and Size Bias

- Failure replay modes feature higher archive hardness (~0.26-0.35) and positive size bias (~0.09-0.12).
- Non-failure modes have moderate hardness (~0.16-0.17) and near-zero size bias.

3. Replay Failure Concentration

- Failure replay modes show increased failure concentration (~0.17-0.33).
- Random and stratified random replay have modest failure concentration (up to ~0.22).
- No replay has zero failure concentration, as expected.

4. Failure Concentration and Transfer

- Higher failure concentration in failure/replay modes correlates with worse transfer performance and higher optimality gaps.
- Best transfer coincides with moderate failure concentration and higher diversity (e.g., stratified random replay).

Code Novelty vs Transfer

- Compression-novelty gating results in **zero code novelty**, yet transfer gaps increase (worsen).
- Conditions with **higher code novelty (0.6-0.9)** correspond to better or more consistent transfer gaps (~5-13%).
- This indicates that **code novelty decrease under compression leads to transfer degradation.**

Specific Replay Mode Classes

Replay Type	Observations
No Replay	Best TSPLIB and transfer gaps, zero failure concentration, moderate archive diversity and hardness.
Random Replay (score_only)	Increased TSPLIB and transfer gaps. Some replay failure concentration (~0.19). Moderate diversity.
Random Replay with Compression (novelty_gate)	Identical worst-case gaps as failure replay modes, zero code novelty. Slight replay selection diversity.

Replay Type	Observations
Stratified Random Replay	Best synthetic gap (0.0), good transfer and TSPLIB gaps. Moderate failure concentration (~0.22).
Diversity-weighted Replay	Higher TSPLIB and transfer gaps than stratified; moderate diversity and failure concentration.
Failure Replay	Highest gaps and failure concentration, lowest archive diversity, positive size bias.
Residual Failure Replay	Similar bad transfer/optimality performance as failure replay, with high failure concentration and low diversity.
Diversity Failure Replay	Moderate transfer/TSPLIB gaps, reduced diversity but better than failure replay.
Diversity Failure Replay with Compression (novelty_gate)	Zero code novelty, transfer gaps worsen compared to score_only.

Mechanism Study Summary

- Failure and residual-failure replay lead to concentrated replay on failures, reducing diversity and worsening transfer gaps.
- Diversity-weighted replay mediates this somewhat but still worse than stratified random or no replay.
- Stratified random replay balances selection diversity and failure concentration better, achieving good performance.
- Compression and novelty gating drastically reduce code novelty, which tracks with worse transfer gaps despite archive descriptor diversity.
- No replay achieves best transfer performance but no improvement in synthetic holdout gap beyond a baseline very close to zero.

****Conservative Interpretation and Recommendations****

- The main evidence shows *****"phase6_no_replay" and "phase6_stratified_random_replay" provide the best holdout transfer performance**** based on TSPLIB gap (~8-9%) and synthetic holdout gap (near zero).
- Failure-based replay modes increase failure concentration and hardness but deteriorate transfer and optimality gaps significantly.
- Compression-driven reduction in code novelty coincides with impaired transfer, indicating ****code novelty is correlated with transfer quality****.
- Stratified random replay improves synthetic holdout gap to zero without sacrificing archive diversity, implying balanced sampling aids transfer.
- Code novelty drops under compression/noulty-gate, but transfer improves only when code novelty is moderate/high; thus, ****code novelty decline co-occurs with transfer degradation under compression****.
- Lexical or archive descriptor diversity alone is insufficient to ensure transfer; ****mechanism design should consider failure concentration and code novelty explicitly.****

Summary

Insight	Evidence Summary
Best transfer accuracy without replay	phase6_no_replay: lowest TSPLIB(8.17%) & transfer gap(5.3%), zero failure concentration
Best synthetic holdout gap with replay	phase6_stratified_random_replay: synthetic gap 0%, TSPLIB gap ~9.1%, transfer gap 5.8%
Failure replay worsens optimality & transfer	Highest TSPLIB gap (~21%), transfer gap (~16%), and failure concentration (~0.26-0.33), reduced diversity
Diversity-weighted replay moderate performance	TSPLIB gap ~15%, transfer gap ~9.7%, moderate failure concentration
Compression reduces code novelty, harms transfer	Compression with novelty gate results in zero code novelty & lower transfer quality despite archive descriptor diversity
Code novelty and transfer correlate	Higher code novelty (~0.6-0.9) coincides with better transfer gaps
Replay failure concentration inversely affects transfer	Higher failure concentration correlates with worse transfer gaps

Final Note

- The best transfer and lowest optimality gaps arise *without failure-based replay* and with *stratified random replay*.
- Compression and novelty-gated replay reduce code novelty, which negatively aligns with transfer improvements.
- Therefore, replay schemes should balance failure coverage, archive diversity, and maintain code novelty to optimize transfer and algorithmic effectiveness.